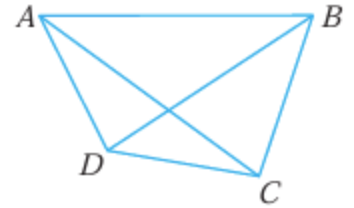


MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.2 Worksheet

1. Use the given diagram to write each combination of vectors as a single vector.



- a) $\overrightarrow{AB} + \overrightarrow{BC}$
- b) $\overrightarrow{CD} + \overrightarrow{DB}$
- c) $\overrightarrow{DB} - \overrightarrow{AB}$
- d) $\overrightarrow{DC} + \overrightarrow{CA} + \overrightarrow{AB}$

2. Copy the vectors in the figure and use them to draw the following vectors.

- a) $\mathbf{a} + \mathbf{b}$



- b) $\mathbf{a} - \mathbf{b}$

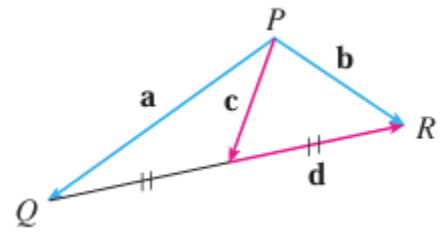
c) $\frac{1}{2}\mathbf{a}$

d) $-3\mathbf{b}$

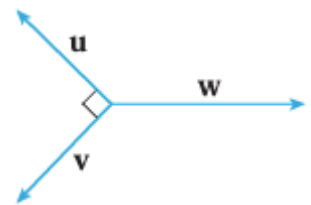
e) $\mathbf{a} + 2\mathbf{b}$

f) $2\mathbf{b} - \mathbf{a}$

3. In the figure, the tip of $\mathbf{c}$ and the tail of $\mathbf{d}$ are both the midpoint of QR . Express $\mathbf{c}$ and $\mathbf{d}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.



4. If the vectors in the figure satisfy $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$ and $\mathbf{u} + \mathbf{v} + \mathbf{w} = \mathbf{0}$, what is $\|\mathbf{w}\|$?



5. Find $\mathbf{a} + \mathbf{b}$, $2\mathbf{a} + 3\mathbf{b}$, $\|\mathbf{a}\|$, and $\|\mathbf{a} - \mathbf{b}\|$ for each of the following sets of vectors $\mathbf{a}$ and $\mathbf{b}$.
- a) $\mathbf{a} = \langle 5, -12 \rangle$ & $\mathbf{b} = \langle -3, -6 \rangle$

b) $\mathbf{a} = 4\mathbf{i} + \mathbf{j}$ & $\mathbf{b} = \mathbf{i} - 2\mathbf{j}$

c) $\mathbf{a} = \langle 1, 2, -3 \rangle$ & $\mathbf{b} = \langle -2, -1, 5 \rangle$

d) $\mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$ & $\mathbf{b} = 2\mathbf{j} - \mathbf{k}$

6. Find a unit vector that has the same direction as the given vector.

a) $\langle -3, 7 \rangle$

b) $-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

7. Find a vector that has the same direction as $\langle -2, 4, 2 \rangle$ but has a length of 6.